

Presentation 2

Link to presentation:

https://github.com/arvindkrishna87/STAT390_Winter2026/blob/master/Team2/Presentations/Slides/Presentation2_Team2.pdf

Link to code:

https://github.com/arvindkrishna87/STAT390_Winter2026/tree/master/Team2/Code

- *Note: our full code for this week has not been committed and uploaded here to Git yet due to issues explained in unresolved issues, which was communicated to our TA Noah*

What did each of the team members do?

Adithi & Jason:

This week, we focused on diagnosing why the model misclassifies cases by systematically checking whether misclassification patterns track with anatomical region, laterality, and other demographic variables. We started by compiling and re-reviewing prior Fall 2025 model outputs across multiple train–test splits and by region and demographic variables to identify where errors were clustering and any variable association with C-MIL misclassification (data/graphs used seen [here](#)). Next, we examined histologic slices from both misclassified and correctly classified cases within each region (especially plica, caruncle, bulbar conjunctiva, and limbus) and looked for differences in tissue architecture. Finally, we linked what we saw in tissue architecture (notably differences in goblet-cell–rich epithelium between regions) back to the error patterns to generate concrete hypotheses and recommendations for future stratified evaluation and potential model adjustments.

Roma & Paige:

- We made the mandatory changes to the code:
 - We introduced KimiaNet instead of DenseNet as the pre-trained network.
 - We removed RandomResizedCrop as a data pre-processing step for patches, and replaced it with RandomResize() (with slight randomized zooming for training variation).
 - We removed saturation jitter when creating image transforms for training.
 - We replaced Adam with AdamW when initializing the optimizer.
 - We added persistent_workers=True when creating data loaders for train/val/test splits.
- As a difficult task, we introduced Optuna hyperparameter tuning to the model:

- To do this, we first learned about Optuna as a process and familiarized ourselves with sampler code.
- Used team 6's mathematical portion of Presentation 1 as well as referenced the code to determine reasonable initial ranges for hyperparameters to tune.
- Made copies of main.py to edit for Optuna training:
 - added a modelhyperparams class, a choose_params() function, an objective function(), and edited the main() function to execute the optuna trials and report the results instead of executing the full MIL training.
- Made copies and edited trainer.py and models.py to account for running Optuna.
- Successfully executed the Optuna run, and got initial validation loss and optimized hyperparams printouts.

Charlene:

- With the mandatory changes made by the coders, I analyzed whether or not they had an effect on our model. I created validation/training loss, confusion matrix, attention distributions, deviation plots, and analyzed the top attention patches on correctly and incorrectly identified high-grade and benign cases. With these graphs, I determined the changes to the model and how it impacts the results of the predictions.

How does it help the project?

Adithi & Jason:

- By tracing misclassified cases back to anatomical region and then validating those patterns on histologic review, we identified that benign lesions from goblet-cell–dense regions—especially the caruncle and plica—are systematically challenging, likely because their normal tissue architecture can resemble features the model associates with higher-risk patterns. Our recommendations focus on adjustments that directly address benign misclassification in these regions, such as accounting for region during evaluation/training and reducing the model's reliance on goblet-cell–heavy visual cues. Because different conjunctival regions have distinct tissue architecture and cell composition, these changes aren't just "fixes" for caruncle/plica—they can also improve robustness and accuracy in other regions by helping the model learn region-appropriate signals instead of over-weighting features that have little association with C-MIL.

Roma & Paige:

- Mandatory changes:
 - KimiaNet is trained on actual histopathology images which better suits the problem which can hopefully increase accuracy. By replacing RandomResizedCrop() with RandomResize(), hope to increase model accuracy because it doesn't change the aspect ratio of the images. Removing saturation jitter can also impact model accuracy as well as AdamW could increase model accuracy because it can achieve better convergence of the loss function.
 - Persistent workers can increase model efficiency due to reducing CPU "dead time" between epochs.
- Optuna hyperparameter tuning:
 - Introducing Optuna hyperparameter tuning to the model can help the project by potentially increasing model accuracy. It allows us to search for optimal hyperparams when we have been unable to practically use grid-search to do so. Can either validate or improve the hyperparam sets we have been using in the model thus far. In the future, it would allow us to more check different hyperparam values that perhaps the mathematicians suggest, and with fine tuning of the Optuna code, we can run more extensive trials which would hopefully help maximize the model performance.

Charlene:

- These analyses provide insight into the model's current limitations and guide the next steps for improvement. Based on the visualizations, we can identify areas where the model struggles, including misclassifying benign patches or over-relying on certain attention regions. This allows us to determine what to focus on for next steps.

Issues faced that haven't been resolved:

Adithi & Jason:

- How to account for the very small sample size of Caruncle and Plica regions and certain ethnicities if we end up stratifying the dataset

Roma & Paige:

- On 2/15, all of our new files with the Optuna hyperparameter tuning disappeared from our folder (copies of [main.py](#), [trainer.py](#), [models.py](#) that were edited for Optuna).

- We have screenshots of many of the added code functions for reference, but this prevented our GitHub commit though, and work will need to be redone for next week.
- Hope to discuss how many trials/epochs we should run since it could take days to run and so our initial run was shortened to only 7 trials this week when testing the Optuna code.
- Some hyperparam combinations caused issues with the code structure- so we need to do more debugging to fix how all the hyperparams are being accepted across the different files.

Charlene:

- Some issues that are prevalent include the overprediction of high-grade cases, indicating class imbalance and biased decision boundaries. Benign cases are consistently misclassified, reflecting uncertainty in the model's interpretation of features in these regions.

Proposed next steps:

Adithi & Jason:

- Incorporate region as a factor in the model so that unique cell composition/architecture is treated as expected variation rather than a malignancy cue
 - This should directly reduce benign misclassification in goblet-cell–dense regions and may stabilize performance in other regions too
 - De-emphasize or give less attention weight to goblet cell–driven features (since goblet cell hyperplasia/abundance appears inconsistent across diagnoses and may be contributing to misclassification of benign tissue)

Roma & Paige:

- Expand the Optuna study (more trials and epochs) for more complete search.
- Refine the hyperparameter search space based on initial results.
- Retrain the model using these best hyperparams to evaluate the effect on model performance.
 - Analyze / experiment with specifically the optimized dropout values across the three usages (ex. 0.11 vs. previous 0.3) to assess impact.

Charlene:

- The next steps include identifying the issues of the misclassification in benign cases, including analyzing misleading features, increasing the representation of benign samples, or adjusting the model or division thresholds to improve recognition of these regions.

References

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- Fall 2025 Presentation 4
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